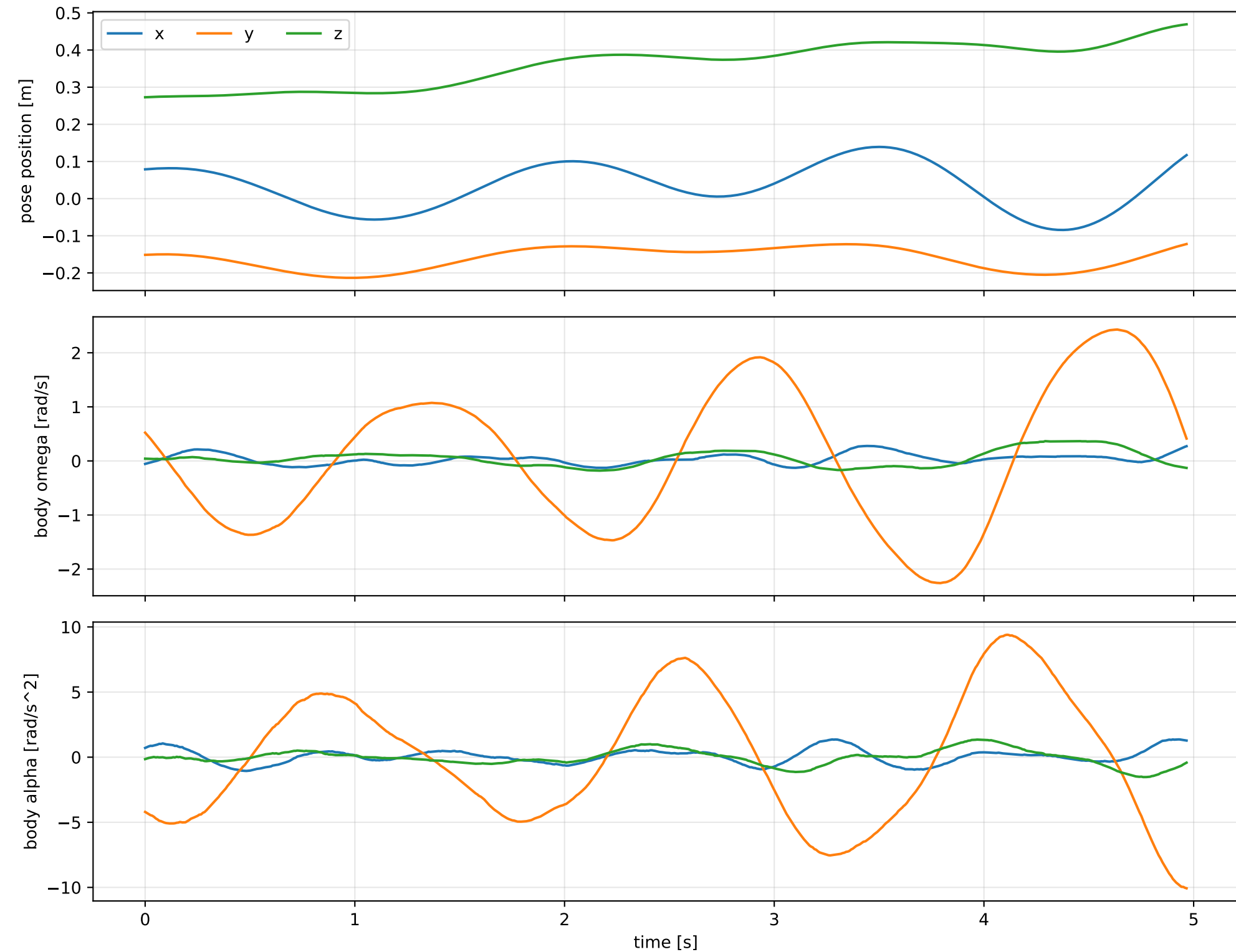
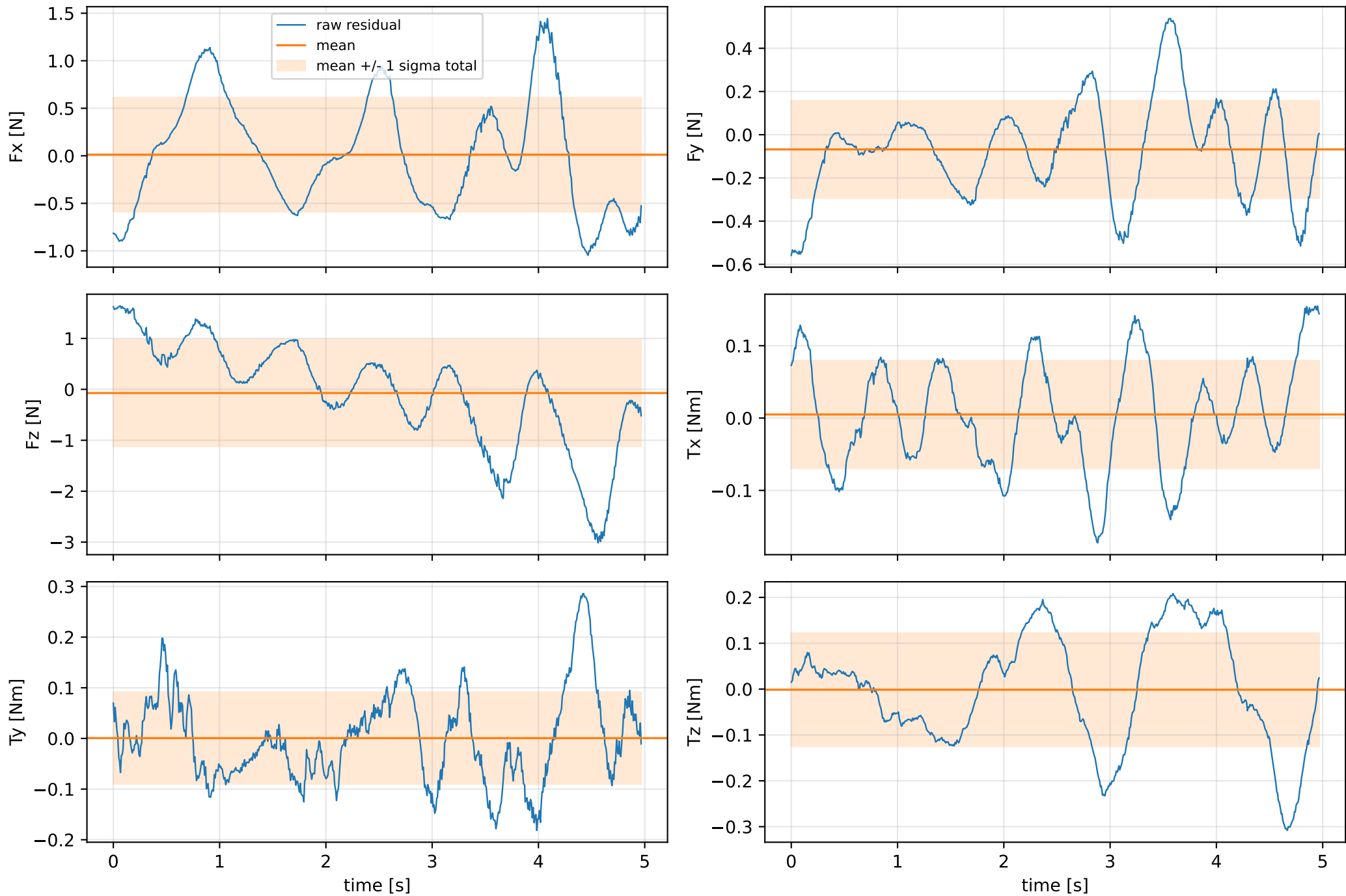


cog_z_nominal: SG trajectory and derived motion



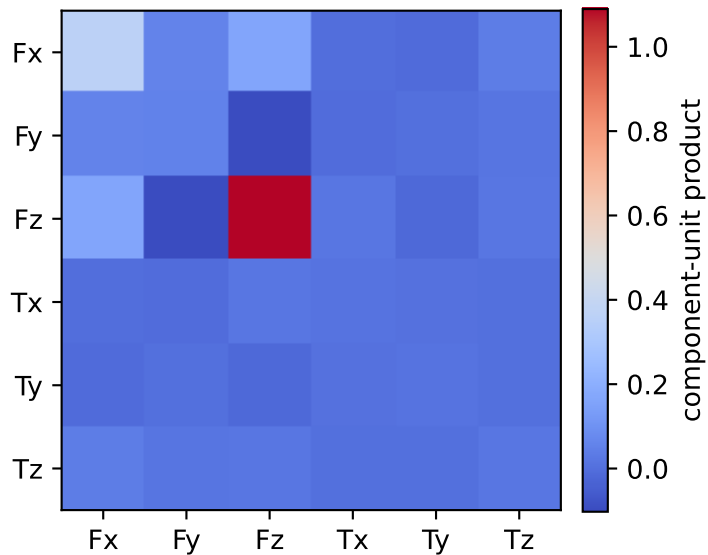
cog_z_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



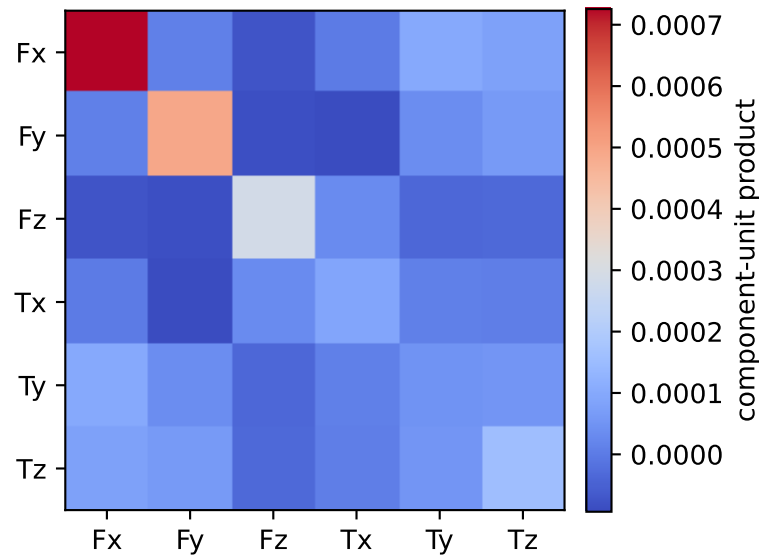
vector RMS about zero: force 1.2278 N, torque 0.17001 Nm; component std about mean: force [0.5998 0.225 1.0438], torque [0.0743 0.0905 0.1233]

cog_z_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.002 0.007 0.011 0.027 0.348 1.131]

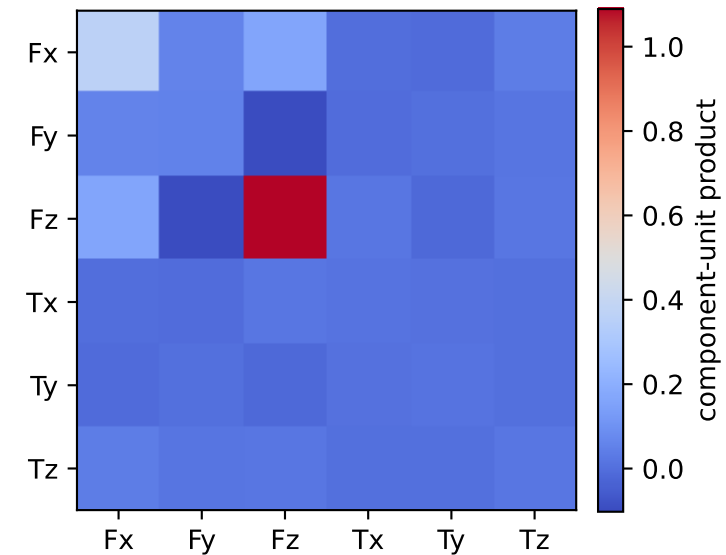
empirical total covariance



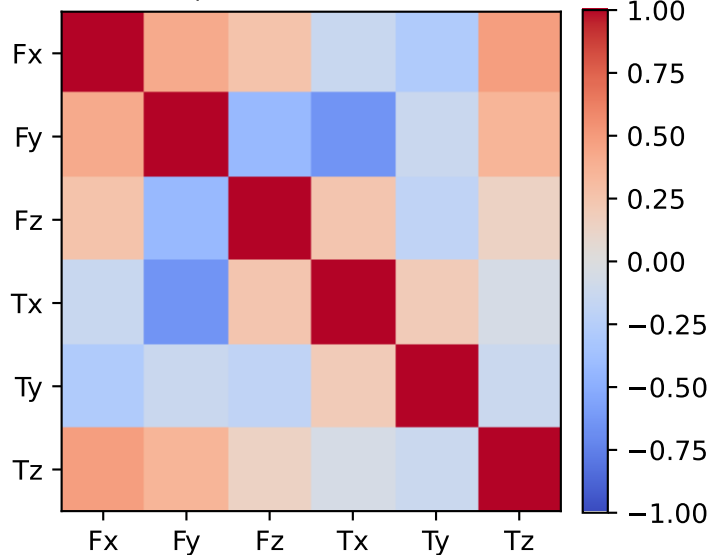
mean full-SG prediction covariance



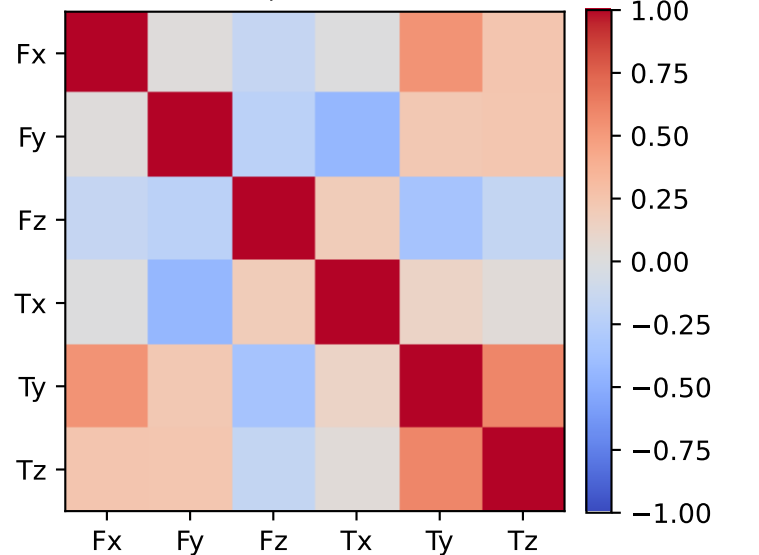
PSD excess model discrepancy covariance



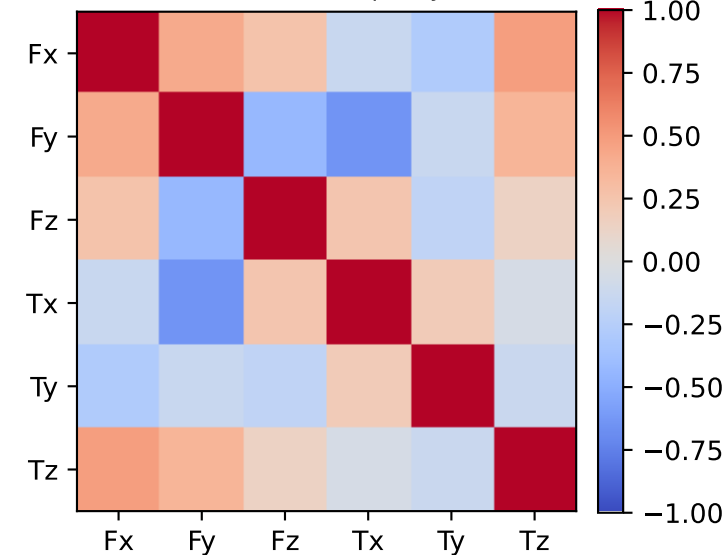
empirical total correlation



mean full-SG prediction correlation

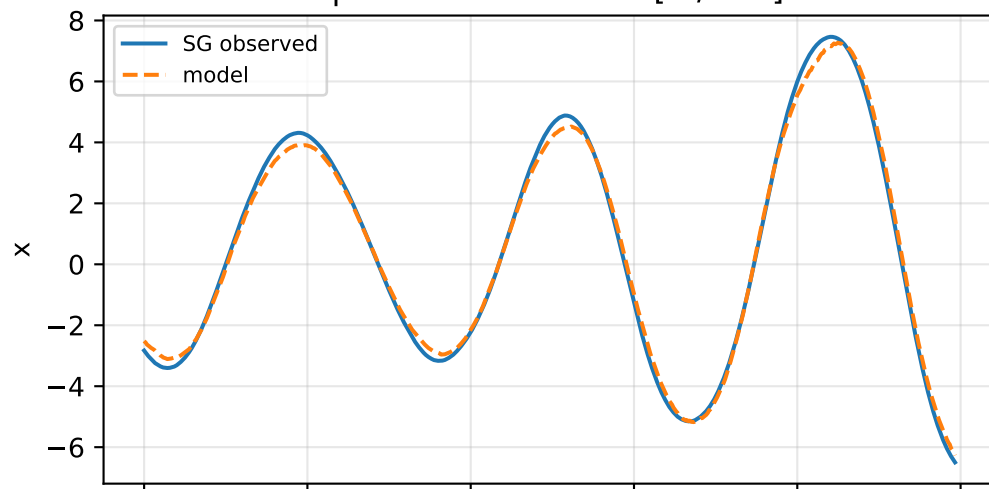


PSD excess model discrepancy correlation

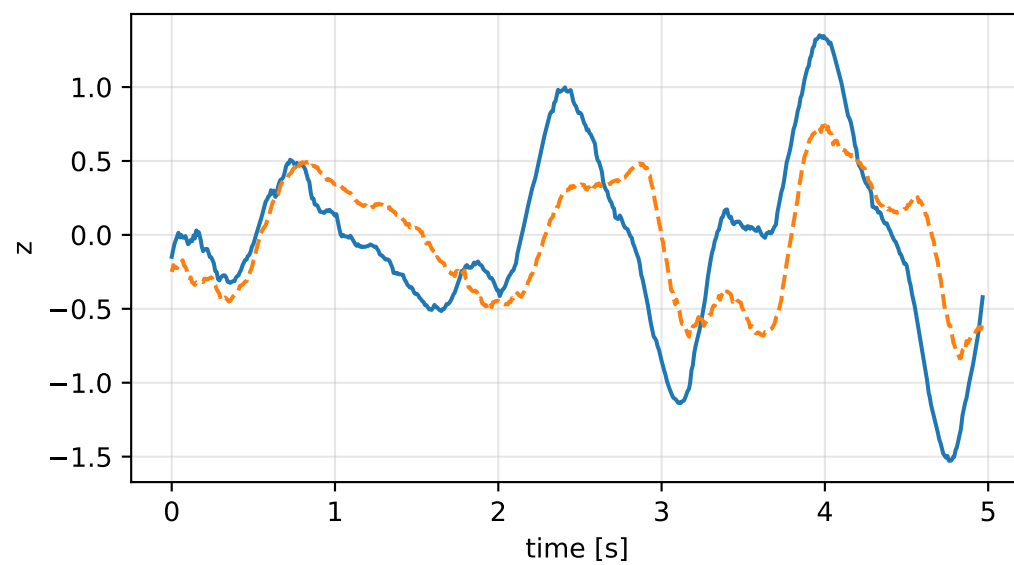
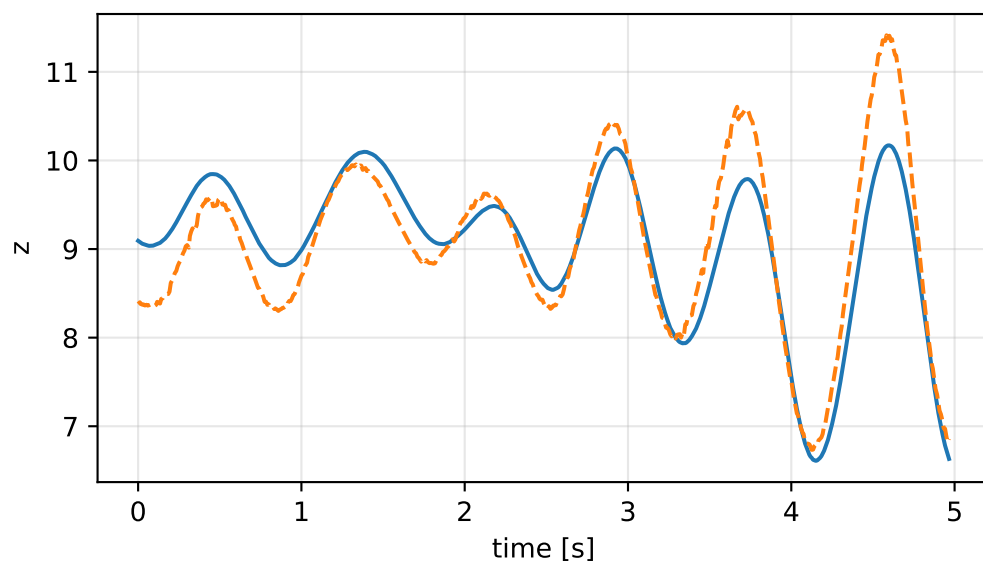
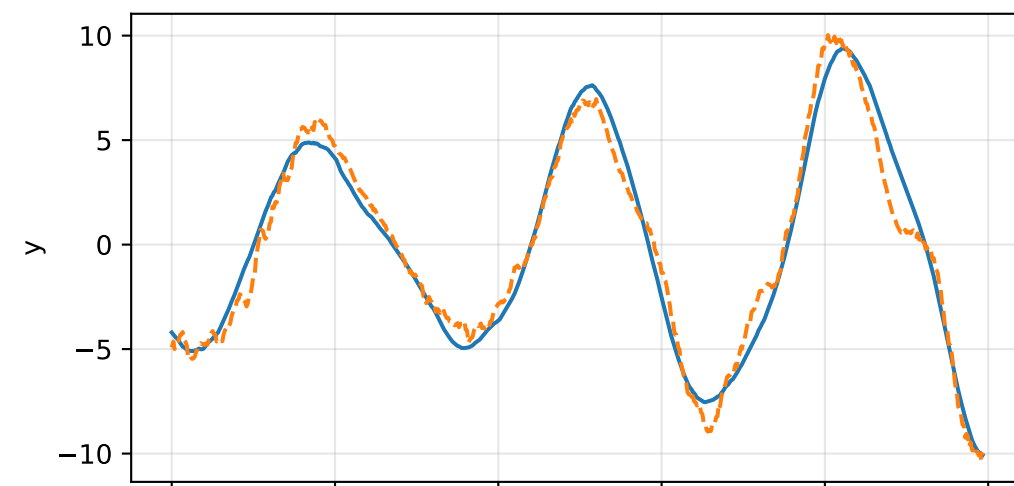
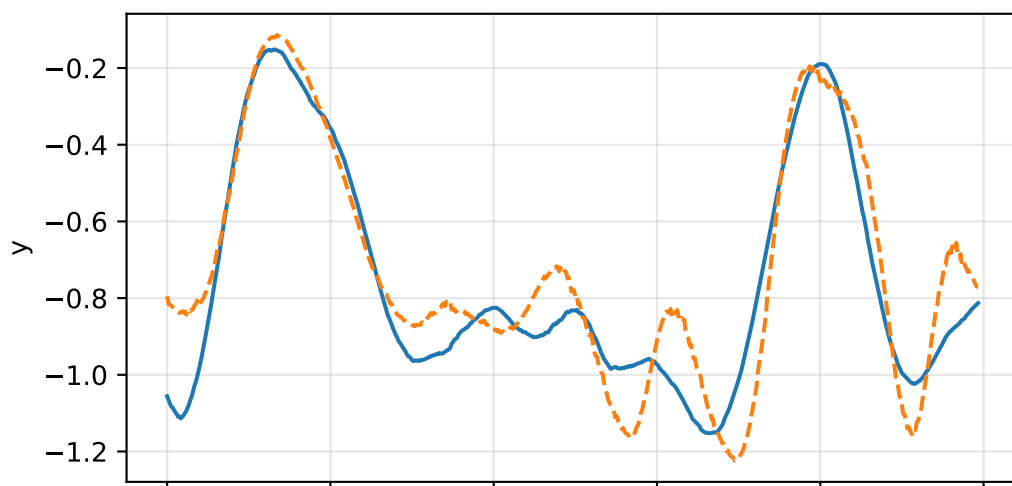
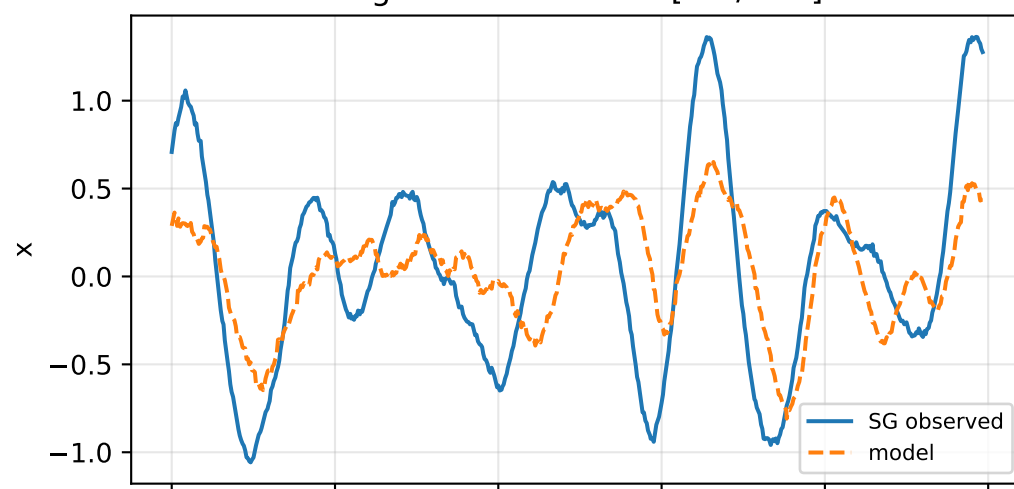


cog_z_nominal: acceleration residual objective

specific acceleration [m/s²]

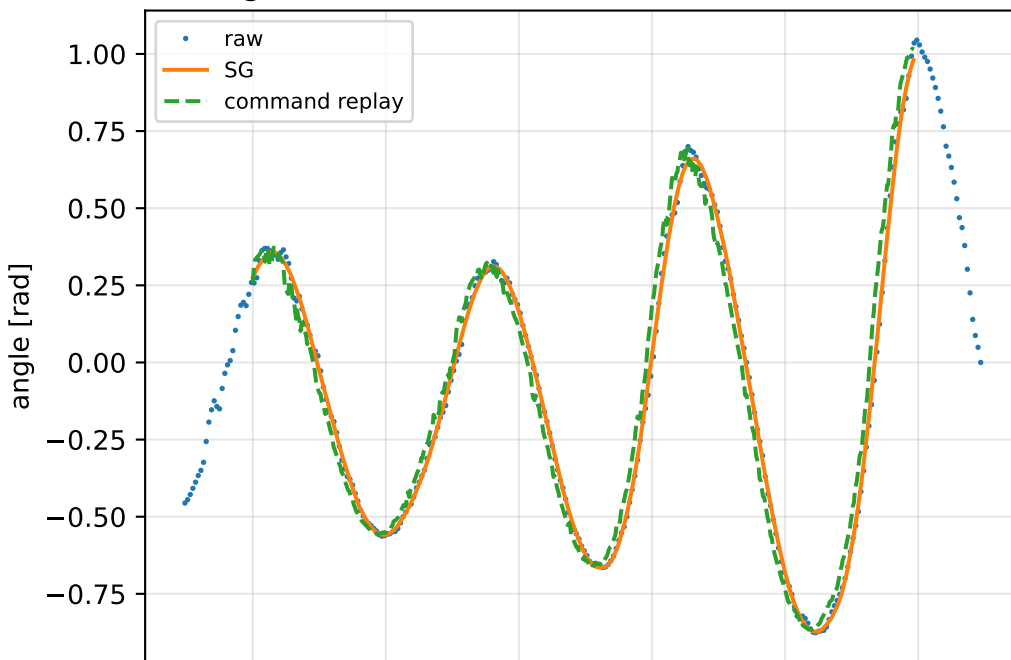


angular acceleration [rad/s²]

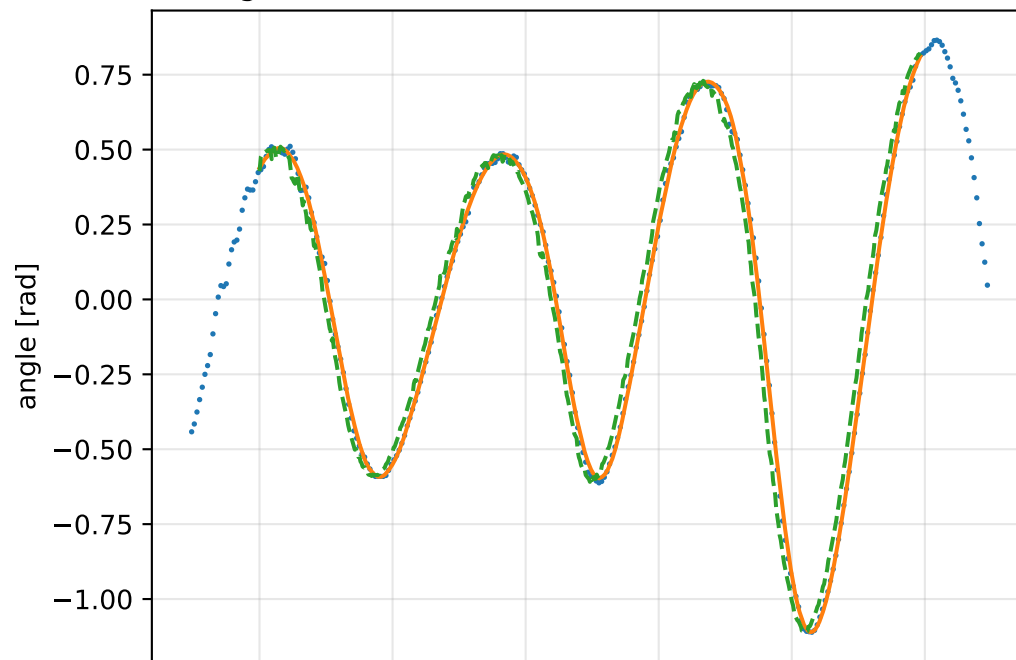


cog_z_nominal: gimbal measurement audit (objective=measured_sg)

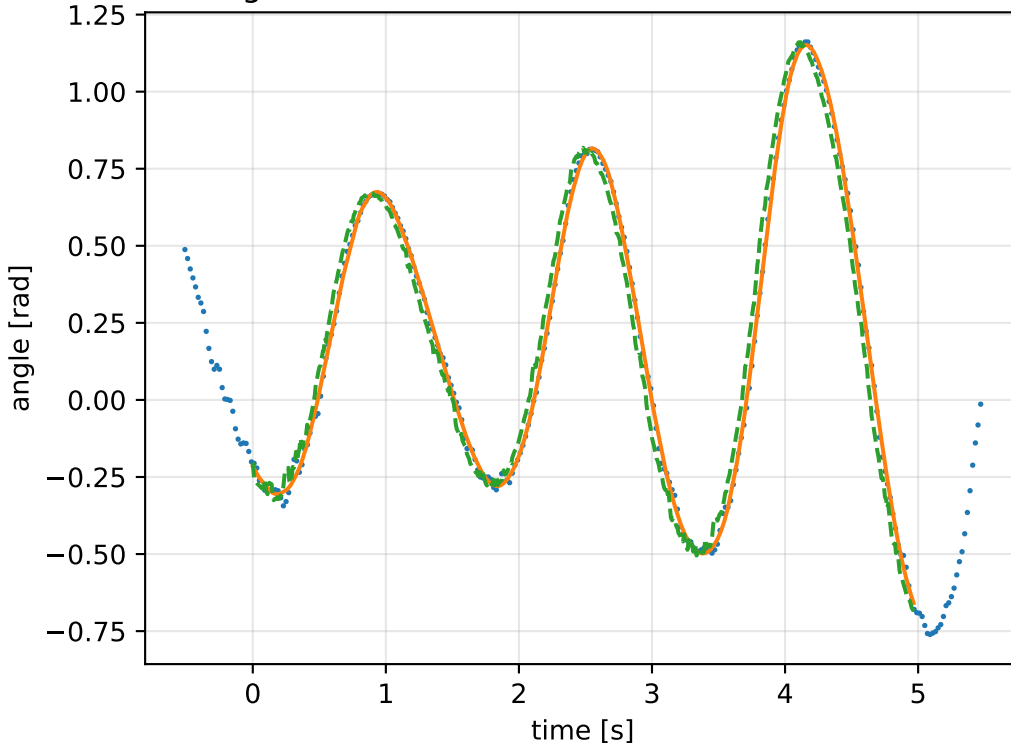
gimbal 1: RMSE=0.0759, max=0.2095 rad



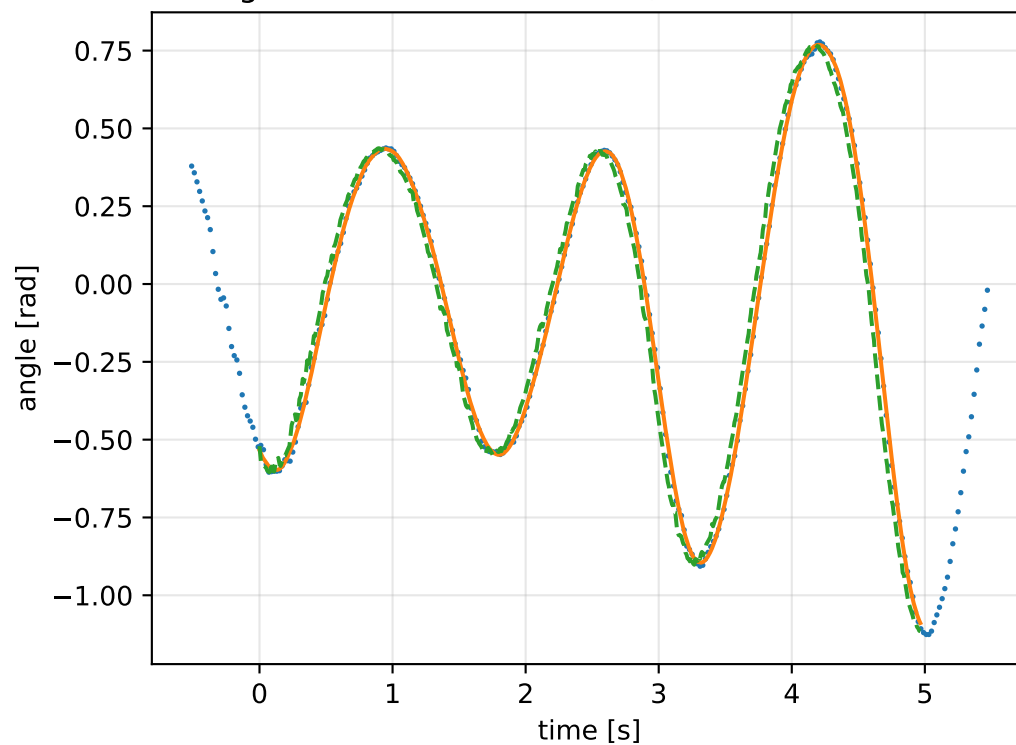
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

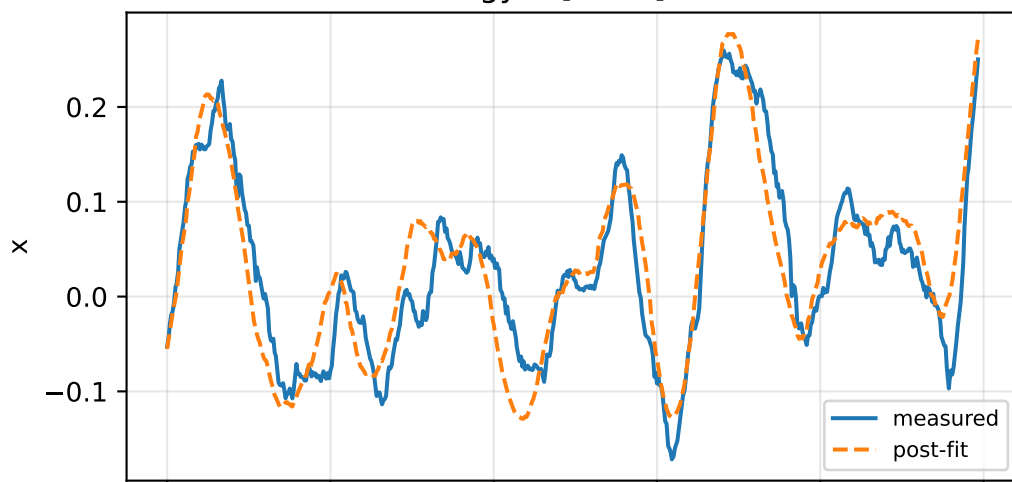


gimbal 4: RMSE=0.07131, max=0.1737 rad

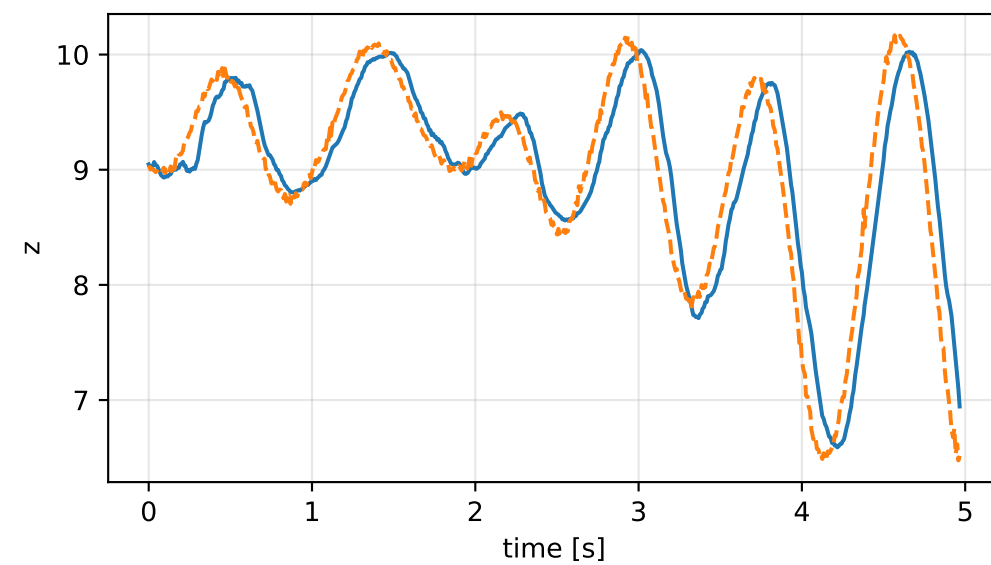
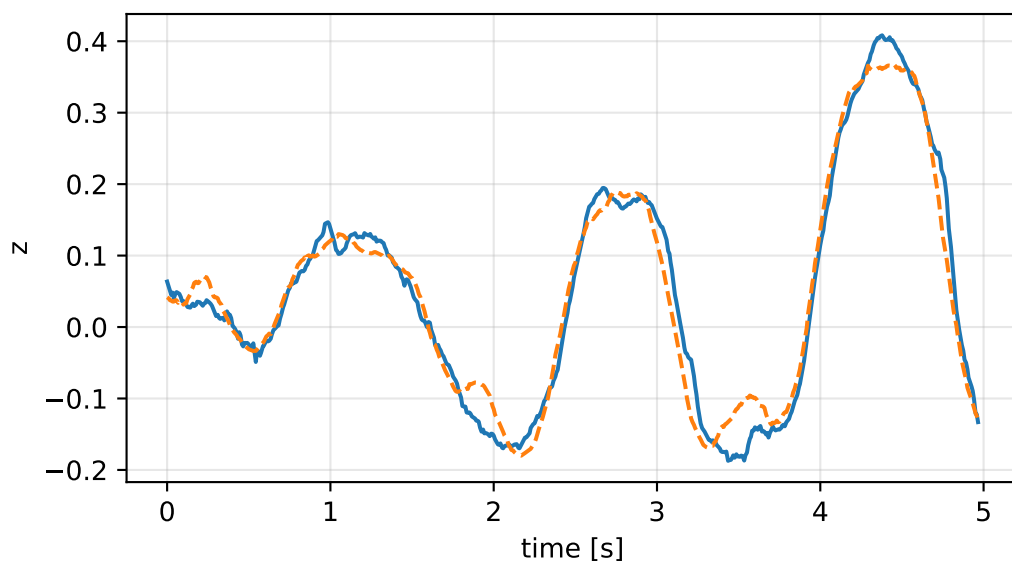
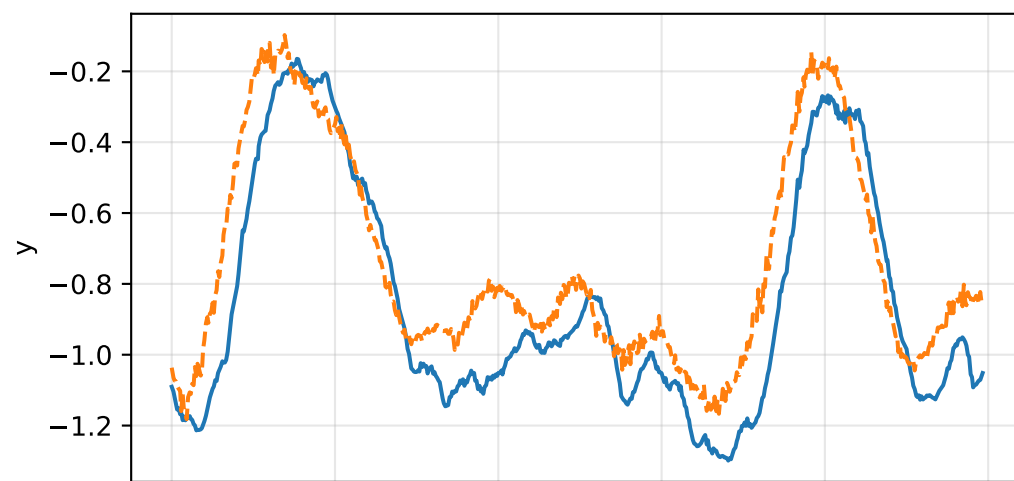
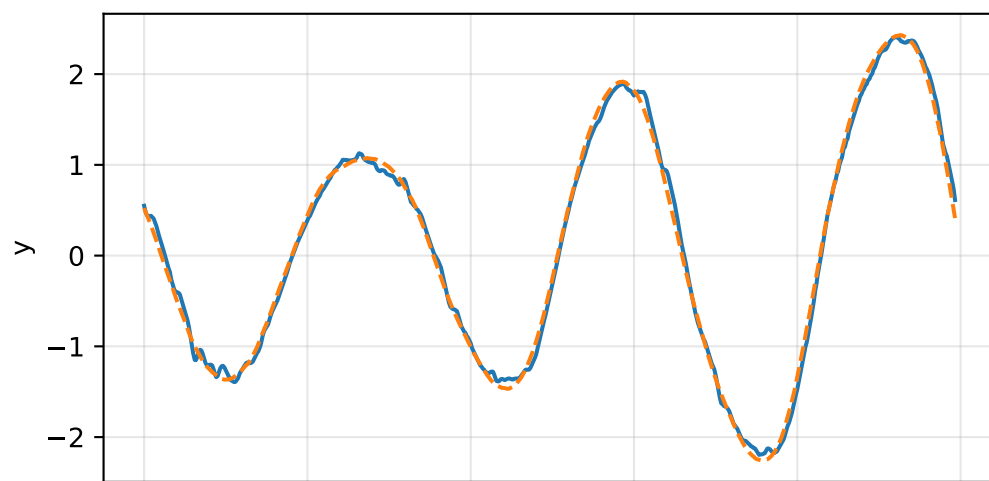
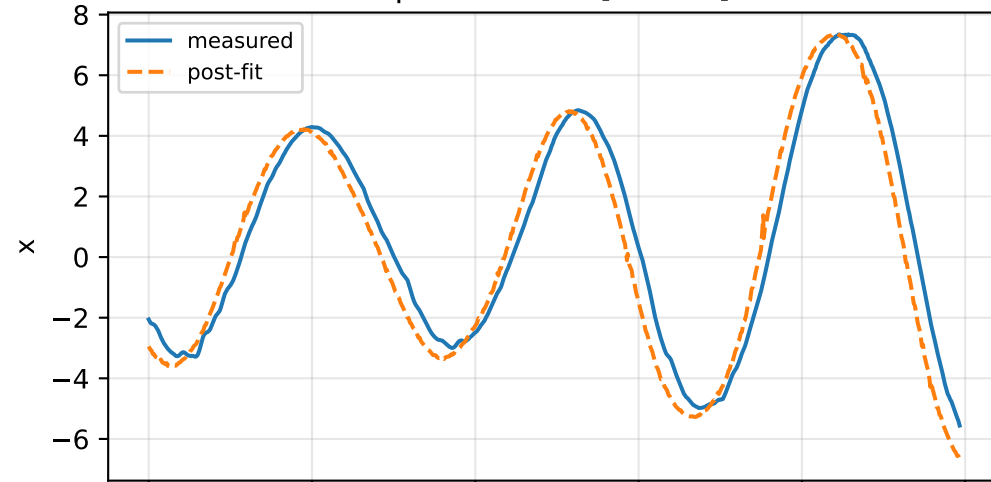


cog_z_nominal: IMU comparison (diagnostic only)

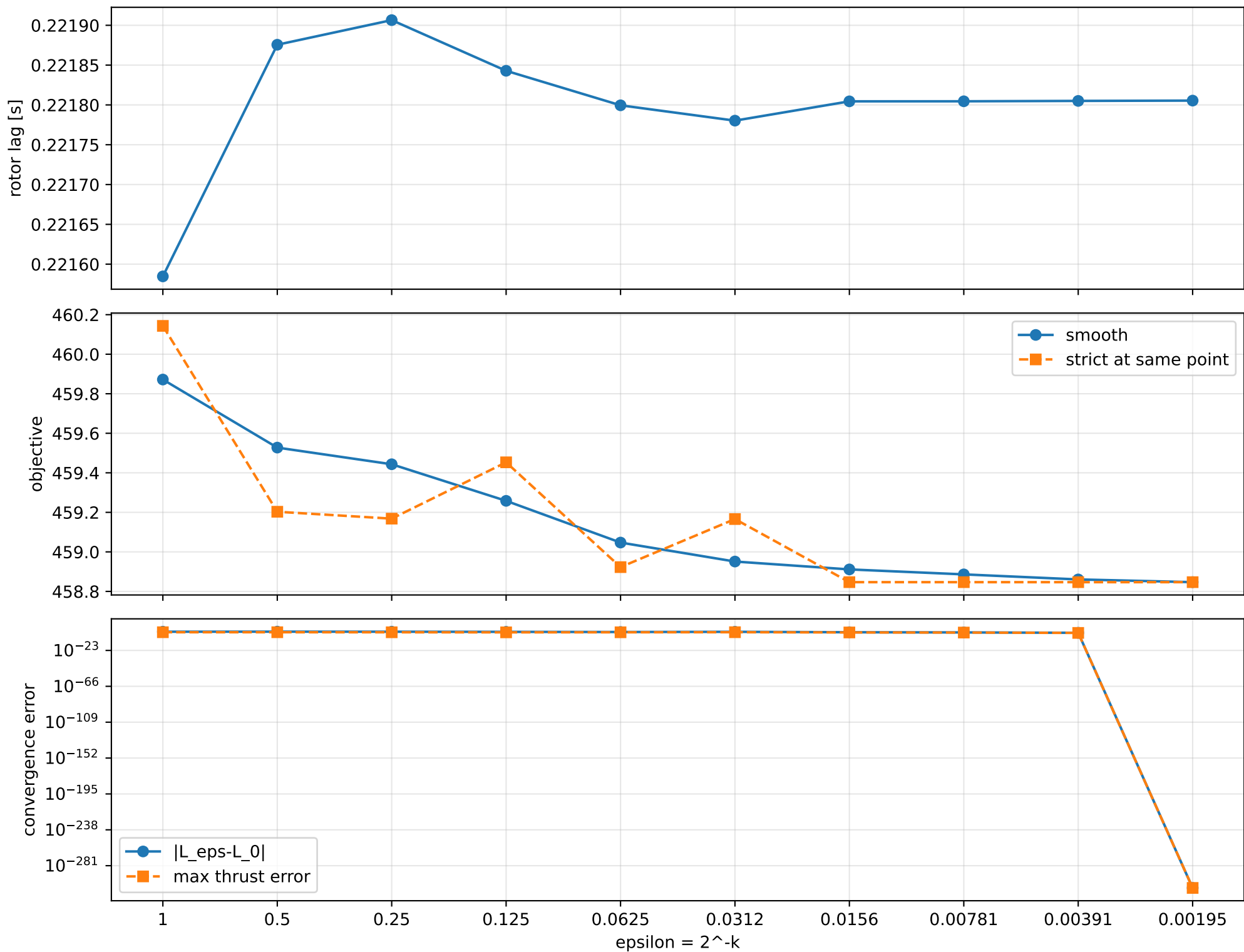
gyro [rad/s]



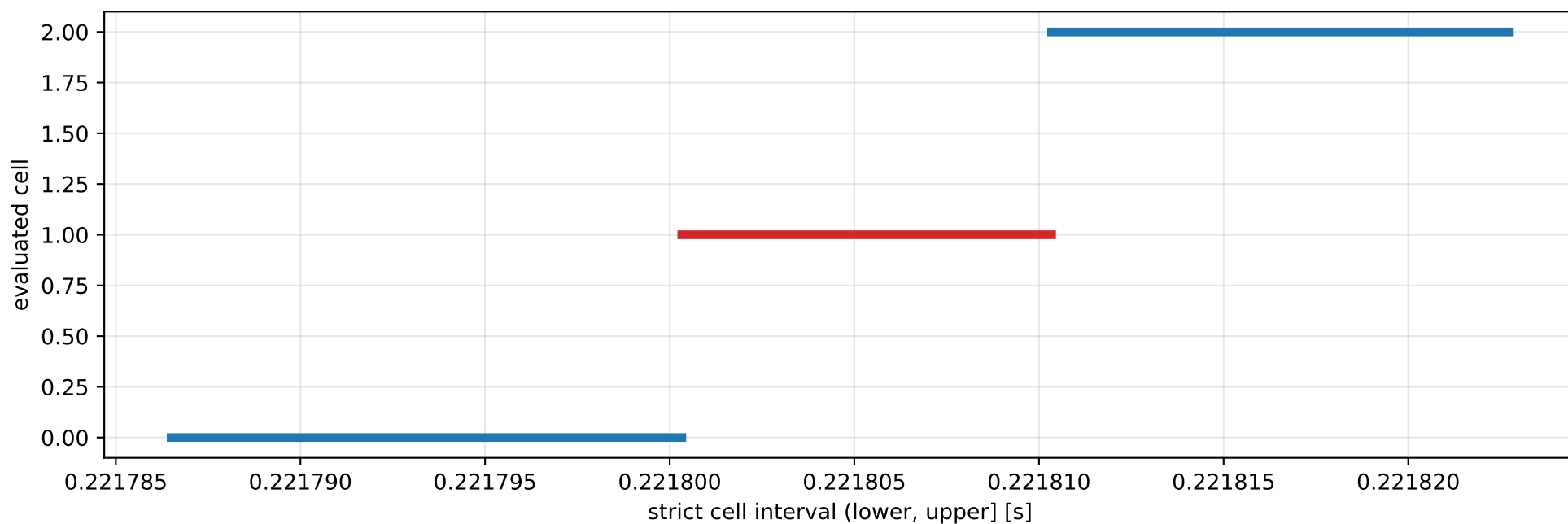
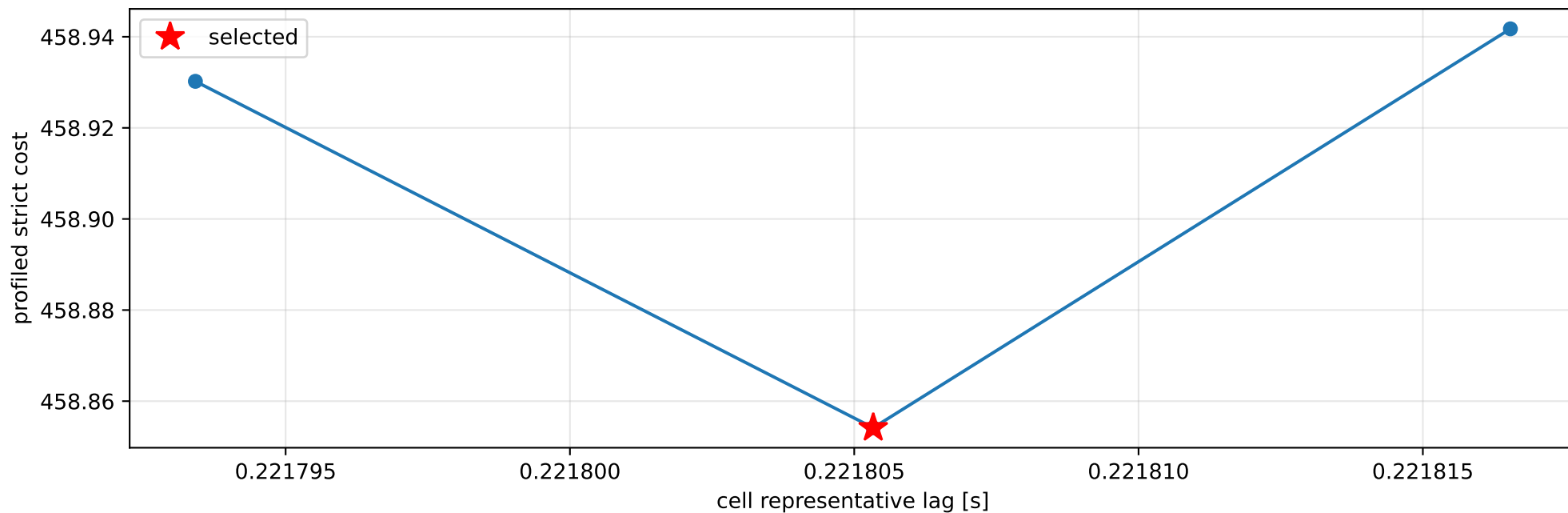
specific force [m/s^2]



cog_z_nominal: smooth-to-strict continuation



cog_z_nominal: exact strict-ZOH lag cells



selected (0.221800327, 0.221810341] s; final smooth 0.221805421 s; data boundary=False

cog_z_nominal: parameter estimate

Case: cog_z_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.31462944

inertia [kg m²]:

```
[[ 0.32158574  0.00297504 -0.03545254]
 [ 0.00297504  0.17751551 -0.00933161]
 [-0.03545254 -0.00933161  0.48365823]]
```

CoG body [m]: [1.76694313e-02 -4.01724327e-06 9.50853911e-03]

force effectiveness: [1.41051717 1.7875999 1.65541749 1.41848486]

scale-free J/m [m²]:

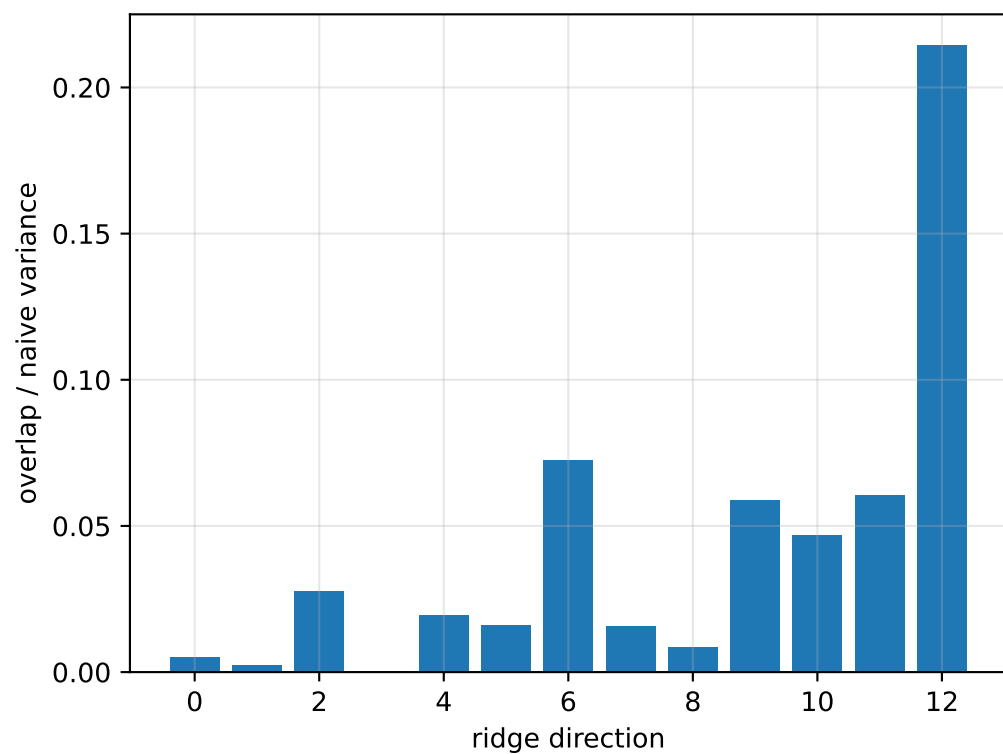
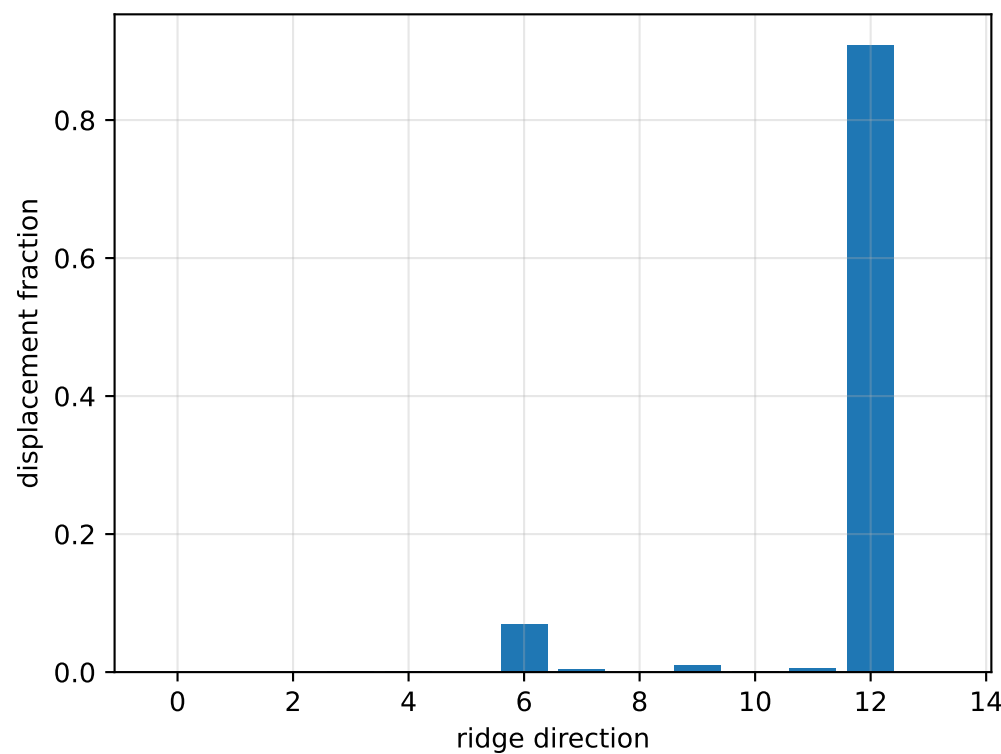
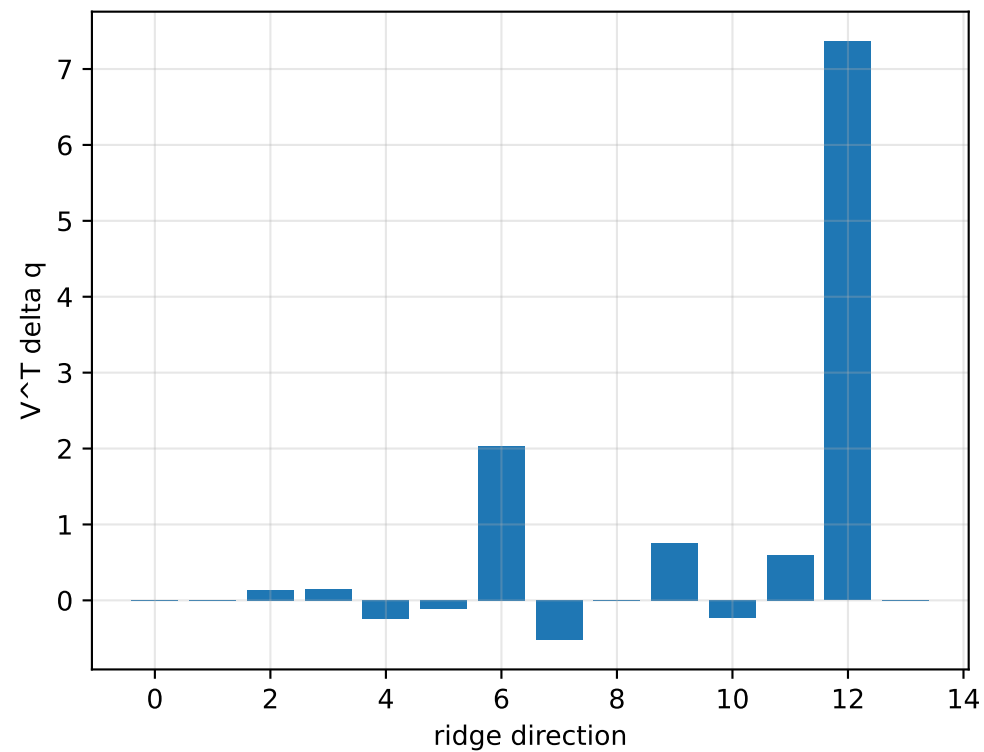
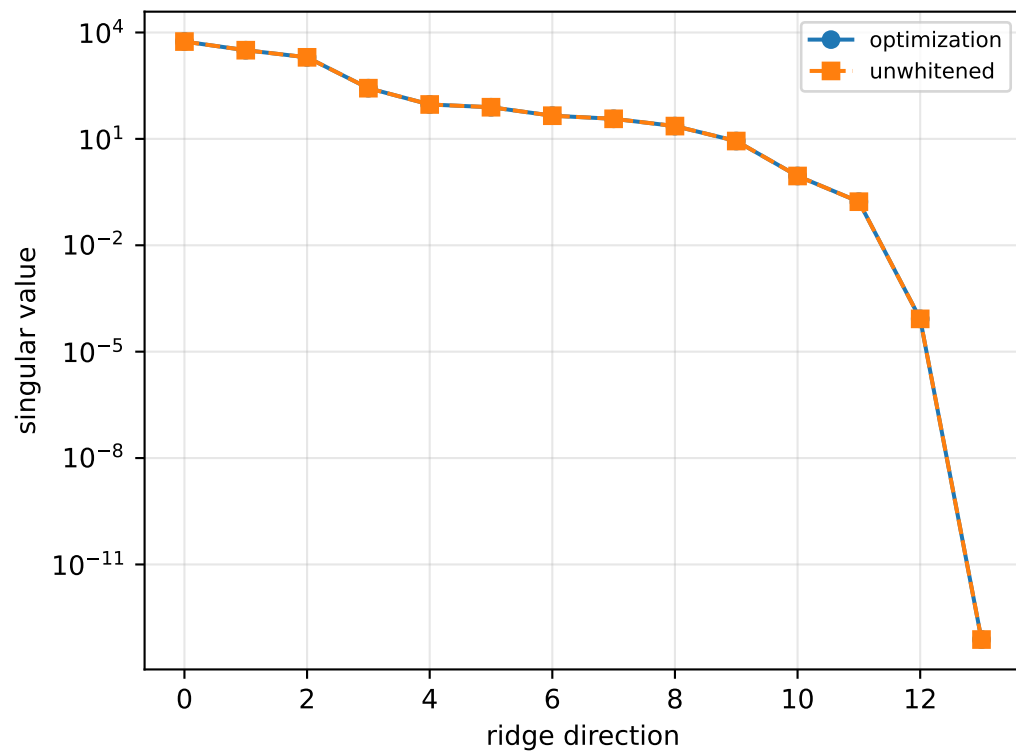
```
[[ 0.0745338  0.00068952 -0.00821682]
 [ 0.00068952  0.0411427  -0.00216278]
 [-0.00821682 -0.00216278  0.11209728]]
```

scale-free f/m: [0.32691502 0.41431134 0.38367547 0.32876169]

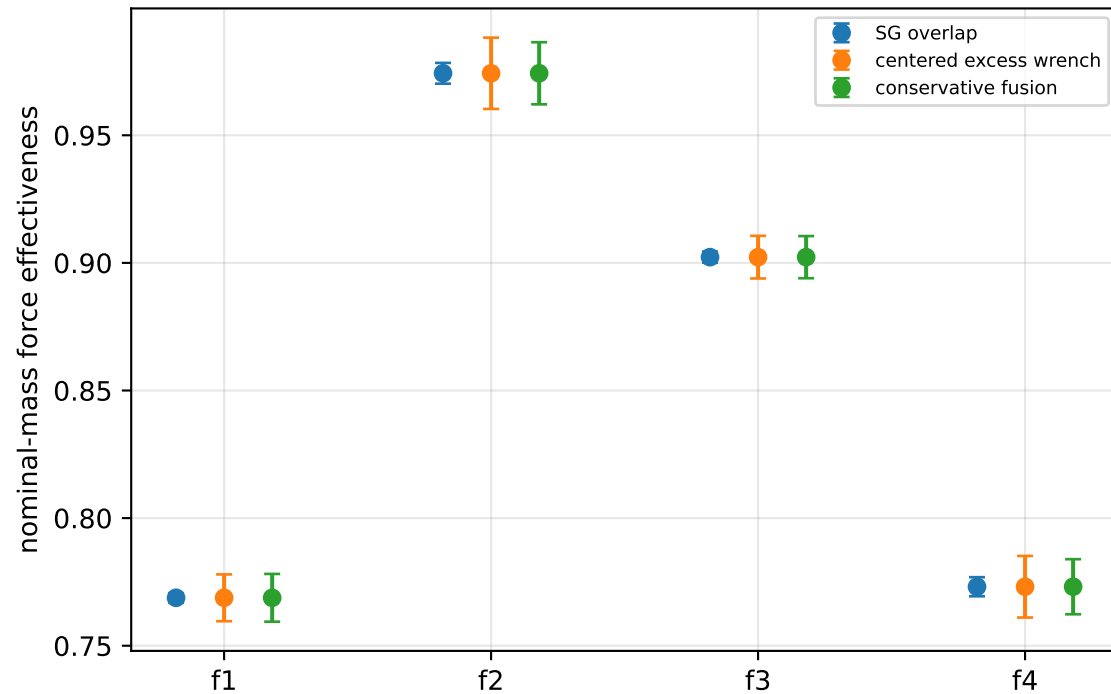
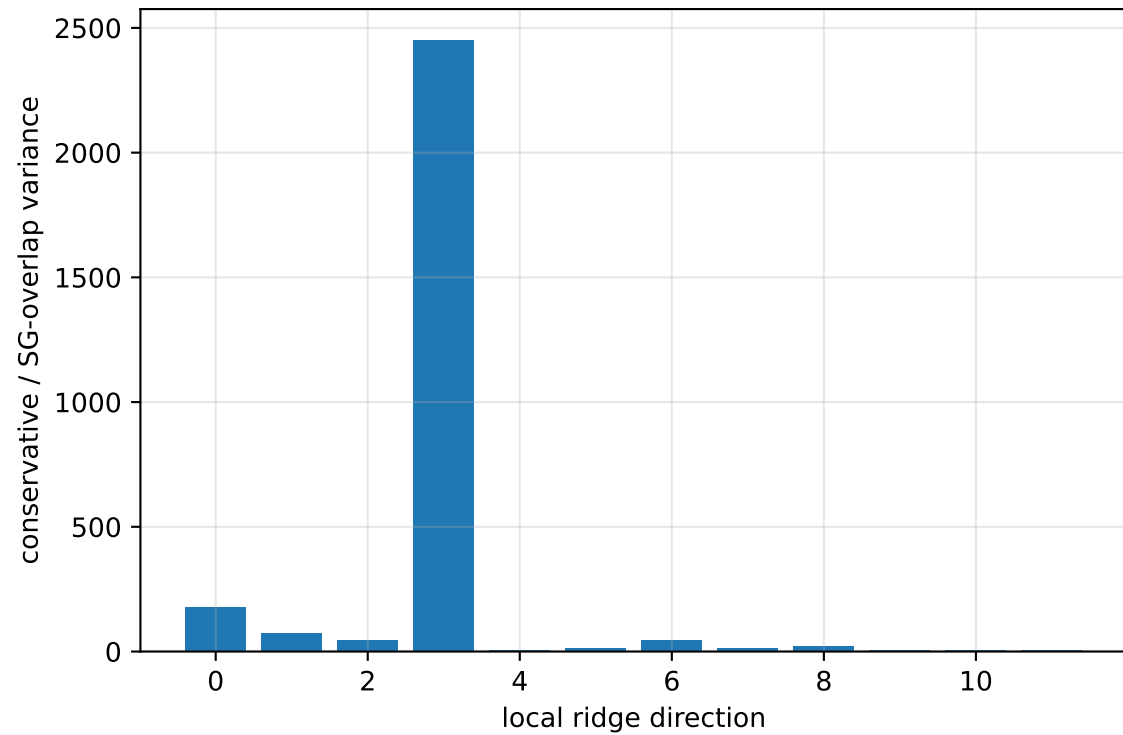
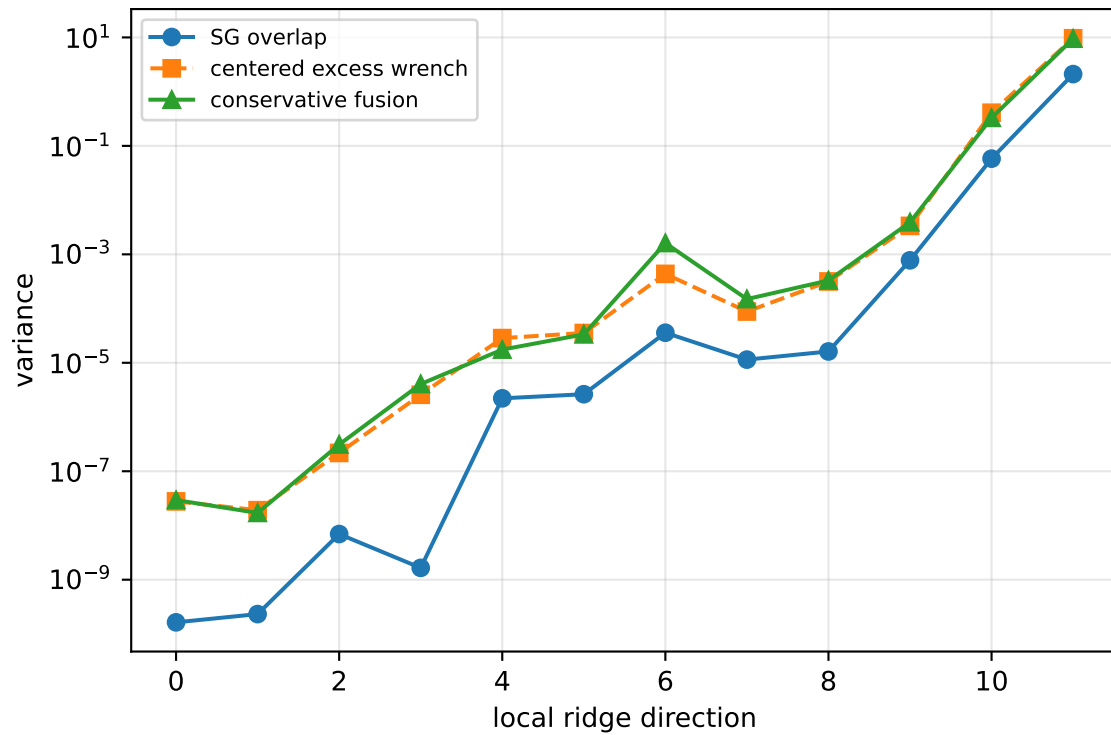
rotor lag cell: (0.221800327, 0.221810341] s

representative: 0.221805334 s

cog_z_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=2.37e-13)



cog_z_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 118.6$
wrench-to-acceleration recovery max error = $4.03\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

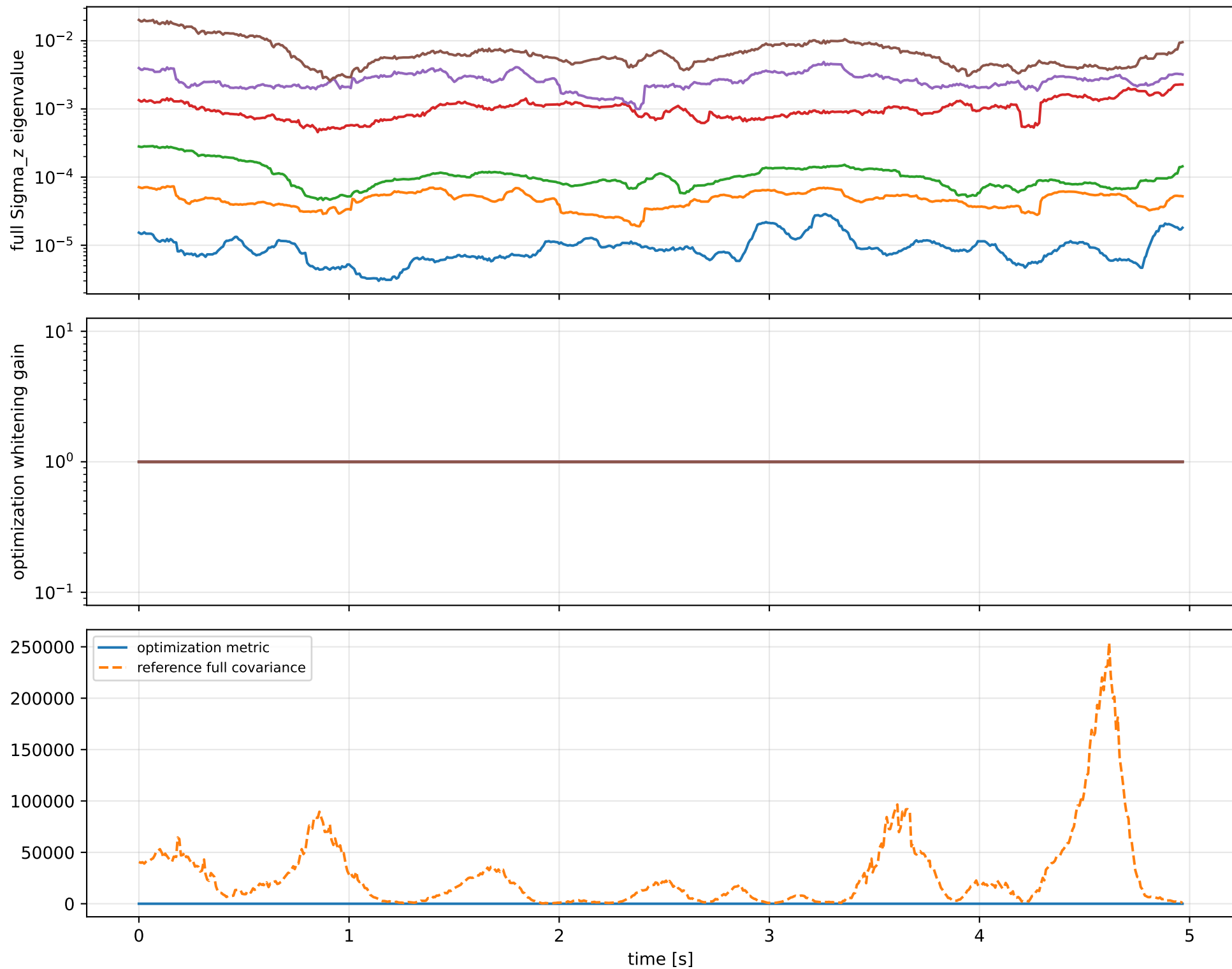
index 11: variance=9.468, ratio=4.492
 $\log_{\text{mass}}=-0.0435$; second_moment_diag_1=-0.462
second_moment_diag_2=+0.0166; second_moment_diag_3=+0.663
second_moment_offdiag_12=+0.201; second_moment_offdiag_13=-0.485
second_moment_offdiag_23=-0.247; cog_x=+6.7e-06
cog_y=-0.000125; cog_z=+1.24e-05
log_force_eff_1=-0.0432; log_force_eff_2=-0.043
log_force_eff_3=-0.0441; log_force_eff_4=-0.044

index 10: variance=0.3288, ratio=5.658
 $\log_{\text{mass}}=+0.0146$; second_moment_diag_1=-0.0308
second_moment_diag_2=+0.208; second_moment_diag_3=-0.248
second_moment_offdiag_12=+0.805; second_moment_offdiag_13=+0.243
second_moment_offdiag_23=-0.431; cog_x=+0.00114
cog_y=-0.00232; cog_z=+9.62e-05
log_force_eff_1=+0.0191; log_force_eff_2=+0.0275
log_force_eff_3=+0.0108; log_force_eff_4=-0.000734

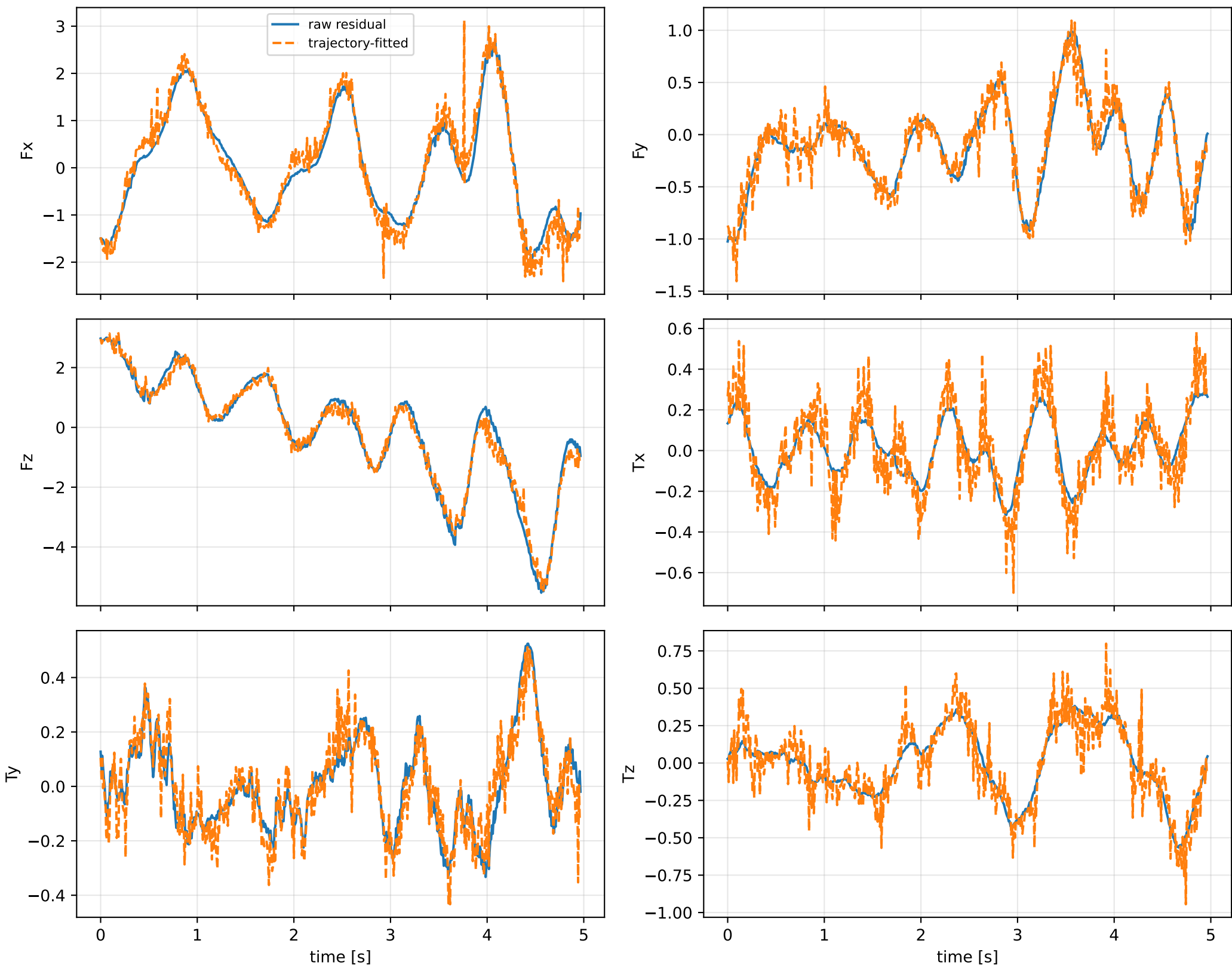
index 9: variance=0.003903, ratio=5.032
 $\log_{\text{mass}}=-0.152$; second_moment_diag_1=-0.0352
second_moment_diag_2=+0.887; second_moment_diag_3=-0.0743
second_moment_offdiag_12=-0.0962; second_moment_offdiag_13=-0.114
second_moment_offdiag_23=+0.208; cog_x=+0.0147
cog_y=-0.00553; cog_z=-0.000601
log_force_eff_1=-0.196; log_force_eff_2=-0.0769
log_force_eff_3=-0.104; log_force_eff_4=-0.249

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_z_nominal: optimization vs reference covariance



cog_z_nominal: wrench / closure (force max 1.73e-14, torque max 5e-16)



cog_z_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_cog_z_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/cog_z_nominal.j

SHA256: 0fc7d4f0f59d87c49bab62593db16eefada27449488e6f94ba9dea5cb570cbdb

data objective: 458.846827209

prior objective: 0.00732647916699

total objective: 458.854153688

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog_z_nominal (cog_position_body_m)

components: z

target: [0.00950975]

std: [1.e-05]

final: [0.00950854]

error: [-1.21049405e-06]

standardized residual: [-0.1210494]

factor objective: 0.00732647916699